

```

=====
N = 10 (h = 0.2000)
=====
t      Exact      Euler Taylor 2nd Midpoint Heun's 3rd
RK4 Adams-Bashforth Adams-Moulton Pred-Corrector
0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
0.20000000 0.19866933 0.20000000 0.20000000 0.19899984 0.19866786
0.19866845 0.19866845 0.19866845 0.19866845 0.19866845
0.40000000 0.38941834 0.39598159 0.39197739 0.38998907 0.38941087
0.38941488 0.38941488 0.38941488 0.38941488 0.38941488
0.60000000 0.56464247 0.57958649 0.56806714 0.56513899 0.56463080
0.56463235 0.56463235 0.56465947 0.56463235
0.80000000 0.71735609 0.74171861 0.72105403 0.71725447 0.71736663
0.71733003 0.71724708 0.71740875 0.71736726
1.00000000 0.84147098 0.87327943 0.84485724 0.84023981 0.84156407
0.84141374 0.84134263 0.84157379 0.84150008
1.20000000 0.93203909 0.96730899 0.93476804 0.92936915 0.93228787
0.93193533 0.93187771 0.93219821 0.93207704
1.40000000 0.98544973 1.02069280 0.98743545 0.98133790 0.98589643
0.98529649 0.98533729 0.98565927 0.98548755
1.60000000 0.99957360 1.03340811 1.00085970 0.99426944 1.00019822
0.99938890 0.99943469 0.99981741 0.99960468
1.80000000 0.97384763 1.00659049 0.97449284 0.96775251 0.97457504
0.97366395 0.97382768 0.97410529 0.97386868
2.00000000 0.90929743 0.94188503 0.90932988 0.90284973 0.91003523
0.90914306 0.90917030 0.90955084 0.90930775
max |error| 0.03526991 0.00369794 0.00644770 0.00073780
0.00018470 0.00016138 0.00025766 0.00003795

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=====
N = 20 (h = 0.1000)
=====
t      Exact      Euler Taylor 2nd Midpoint Heun's 3rd
RK4 Adams-Bashforth Adams-Moulton Pred-Corrector
0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
0.00000000 0.00000000 0.00000000 0.00000000
0.10000000 0.09983342 0.10000000 0.10000000 0.09987501 0.09983338
0.09983340 0.09983340 0.09983340 0.09983340
0.20000000 0.19866933 0.19949992 0.19900025 0.19875086 0.19866916
0.19866929 0.19866929 0.19866929 0.19866929
0.30000000 0.29552021 0.29749670 0.29600773 0.29563567 0.29551975
0.29552010 0.29552010 0.29552047 0.29552010
0.40000000 0.38941834 0.39297822 0.39004702 0.38955517 0.38941752
0.38941814 0.38941483 0.38941901 0.38941855
0.50000000 0.47942554 0.48492088 0.48017126 0.47956353 0.47942440
0.47942517 0.47941905 0.47942674 0.47942607
0.60000000 0.56464247 0.57229585 0.56547325 0.56475437 0.56464141
0.56464185 0.56463341 0.56464438 0.56464331
0.70000000 0.64421769 0.65408742 0.64509622 0.64427149 0.64421752

```

0.64421668	0.64420669	0.64422052	0.64421880
0.80000000	0.71735609	0.72932389	0.71824366
0.71735451	0.71734385	0.71736005	0.71735743
0.90000000	0.78332691	0.79711598	0.78418766
0.78332455	0.78331413	0.78333217	0.78332841
1.00000000	0.84147098	0.85669373	0.84227545
0.84146763	0.84145830	0.84147766	0.84147257
1.10000000	0.89120736	0.90743147	0.89193450
0.89120280	0.89119529	0.89121550	0.89120896
1.20000000	0.93203909	0.94885447	0.93267654
0.93203318	0.93202800	0.93204866	0.93204064
1.30000000	0.96355819	0.98062847	0.96410111
0.96355092	0.96354833	0.96356908	0.96355963
1.40000000	0.98544973	1.00253898	0.98589879
0.98544122	0.98544121	0.98546178	0.98545103
1.50000000	0.99749499	1.01447019	0.99785420
0.99748549	0.99748783	0.99750799	0.99749611
1.60000000	0.99957360	1.01639011	0.99984844
0.99956348	0.99956778	0.99958733	0.99957452
1.70000000	0.99166481	1.00834423	0.99186081
0.99165451	0.99166027	0.99167903	0.99166552
1.80000000	0.97384763	0.99045579	0.97396949
0.97383759	0.97384431	0.97386211	0.97384813
1.90000000	0.94630009	0.96292929	0.94635111
0.94629067	0.94629796	0.94631462	0.94630037
2.00000000	0.90929743	0.92605401	0.90927926
0.90928891	0.90929647	0.90931183	0.90929750
max error	0.01708925	0.00088757	0.00135751
0.00001030	0.00001278	0.00001453	0.00000160

=====

N = 40 (h = 0.0500)

=====

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd
RK4	Adams-Bashforth	Adams-Moulton	Pred-Corrector		
0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
0.05000000	0.04997917	0.05000000	0.05000000	0.04998438	0.04997917
0.04997917	0.04997917	0.04997917	0.04997917	0.04997917	0.04997917
0.10000000	0.09983342	0.09993751	0.09987502	0.09984380	0.09983341
0.09983342	0.09983342	0.09983342	0.09983342	0.09983342	0.09983342
0.15000000	0.14943813	0.14968756	0.14950035	0.14945358	0.14943812
0.14943813	0.14943813	0.14943814	0.14943814	0.14943813	0.14943813
0.20000000	0.19866933	0.19912527	0.19875183	0.19868961	0.19866931
0.19866933	0.19866922	0.19866935	0.19866935	0.19866934	0.19866933
0.25000000	0.24740396	0.24812590	0.24750621	0.24742868	0.24740392
0.24740396	0.24740374	0.24740399	0.24740399	0.24740397	0.24740396
0.30000000	0.29552021	0.29656487	0.29564146	0.29554879	0.29552015
0.29552020	0.29551989	0.29552024	0.29552024	0.29552023	0.29552020
0.35000000	0.34289781	0.34431796	0.34303705	0.34292944	0.34289773

0.34289780	0.34289739	0.34289786	0.34289784
0.40000000	0.38941834	0.39126145	0.38957429
0.38941833	0.38941783	0.38941840	0.38941838
0.45000000	0.43496553	0.43727237	0.43513664
0.43496552	0.43496493	0.43496561	0.43496558
0.50000000	0.47942554	0.48222891	0.47960999
0.47942552	0.47942486	0.47942563	0.47942559
0.55000000	0.52268723	0.52601082	0.52288301
0.52268720	0.52268648	0.52268734	0.52268729
0.60000000	0.56464247	0.56849998	0.56484736
0.56464244	0.56464166	0.56464261	0.56464254
0.65000000	0.60518641	0.60958102	0.60539805
0.60518636	0.60518554	0.60518657	0.60518648
0.70000000	0.64421769	0.64914202	0.64443368
0.64421763	0.64421678	0.64421788	0.64421776
0.75000000	0.68163876	0.68707523	0.68185669
0.68163868	0.68163783	0.68163898	0.68163884
0.80000000	0.71735609	0.72327775	0.71757361
0.71735599	0.71735515	0.71735635	0.71735617
0.85000000	0.75128041	0.75765221	0.75149528
0.75128029	0.75127946	0.75128070	0.75128049
0.90000000	0.78332691	0.79010732	0.78353711
0.78332677	0.78332597	0.78332724	0.78332700
0.95000000	0.81341550	0.82055833	0.81361923
0.81341533	0.81341458	0.81341588	0.81341559
1.00000000	0.84147098	0.84892734	0.84166667
0.84147078	0.84147009	0.84147140	0.84147107
1.05000000	0.86742323	0.87514347	0.86760959
0.86742299	0.86742236	0.86742368	0.86742331
1.10000000	0.89120736	0.89914291	0.89138341
0.89120709	0.89120654	0.89120786	0.89120744
1.15000000	0.91276394	0.92086885	0.91292895
0.91276363	0.91276316	0.91276448	0.91276402
1.20000000	0.93203909	0.94027133	0.93219257
0.93203874	0.93203835	0.93203967	0.93203916
1.25000000	0.94898462	0.95730701	0.94912632
0.94898423	0.94898393	0.94898524	0.94898469
1.30000000	0.96355819	0.97193901	0.96368804
0.96355776	0.96355755	0.96355884	0.96355825
1.35000000	0.97572336	0.98413660	0.97584145
0.97572289	0.97572277	0.97572405	0.97572342
1.40000000	0.98544973	0.99387509	0.98555628
0.98544923	0.98544920	0.98545045	0.98544979
1.45000000	0.99271299	1.00113563	0.99280830
0.99271246	0.99271251	0.99271374	0.99271304
1.50000000	0.99749499	1.00590512	0.99757941
0.99749443	0.99749455	0.99749576	0.99749503
1.55000000	0.99978376	1.00817616	0.99985770
0.99978319	0.99978338	0.99978456	0.99978381
1.60000000	0.99957360	1.00794700	0.99963744
			0.99930340
			0.99958181

0.99957301	0.99957326	0.99957442	0.99957364
1.65000000	0.99686503	1.00522154	0.99691916
0.99686443	0.99686473	0.99686586	0.99686506
1.70000000	0.99166481	1.00000939	0.99170960
0.99166421	0.99166456	0.99166565	0.99166484
1.75000000	0.98398595	0.99232587	0.98402172
0.98398535	0.98398574	0.98398680	0.98398597
1.80000000	0.97384763	0.98219203	0.97387466
0.97384704	0.97384746	0.97384849	0.97384765
1.85000000	0.96127520	0.96963468	0.96129372
0.96127463	0.96127508	0.96127607	0.96127521
1.90000000	0.94630009	0.95468638	0.94631026
0.94629954	0.94630000	0.94630095	0.94630009
1.95000000	0.92895972	0.93738542	0.92896166
0.92895919	0.92895967	0.92896058	0.92895972
2.00000000	0.90929743	0.91777579	0.90929119
0.90929693	0.90929742	0.90929828	0.90929742
max error	0.00847836	0.00021793	0.00031370
0.00000060	0.00000095	0.00000086	0.00000009

Error table | $h = 0.2$ ($N = 10$)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	0.000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00	0.00e+00	0.00e+00			
0.20000000	0.198669	1.33e-03	1.33e-03	3.31e-04	1.47e-06	8.78e-07
8.78e-07	8.78e-07	8.78e-07	8.78e-07			
0.40000000	0.389418	6.56e-03	2.56e-03	5.71e-04	7.47e-06	3.46e-06
3.46e-06	3.46e-06	3.46e-06	3.46e-06			
0.60000000	0.564642	1.49e-02	3.42e-03	4.97e-04	1.17e-05	1.01e-05
1.01e-05	1.70e-05	1.01e-05	1.01e-05			
0.80000000	0.717356	2.44e-02	3.70e-03	1.02e-04	1.05e-05	2.61e-05
1.09e-04	5.27e-05	1.12e-05	1.12e-05			
1.00000000	0.841471	3.18e-02	3.39e-03	1.23e-03	9.31e-05	5.72e-05
1.28e-04	1.03e-04	2.91e-05	2.91e-05			
1.20000000	0.932039	3.53e-02	2.73e-03	2.67e-03	2.49e-04	1.04e-04
1.61e-04	1.59e-04	3.80e-05	3.80e-05			
1.40000000	0.985450	3.52e-02	1.99e-03	4.11e-03	4.47e-04	1.53e-04
1.12e-04	2.10e-04	3.78e-05	3.78e-05			
1.60000000	0.999574	3.38e-02	1.29e-03	5.30e-03	6.25e-04	1.85e-04
1.39e-04	2.44e-04	3.11e-05	3.11e-05			
1.80000000	0.973848	3.27e-02	6.45e-04	6.10e-03	7.27e-04	1.84e-04
2.00e-05	2.58e-04	2.10e-05	2.10e-05			
2.00000000	0.909297	3.26e-02	3.25e-05	6.45e-03	7.38e-04	1.54e-04
1.27e-04	2.53e-04	1.03e-05	1.03e-05			

Error table | $h = 0.1$ ($N = 20$)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	0.000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00		0.00e+00			
0.10000000	0.099833	1.67e-04	1.67e-04	4.16e-05	3.39e-08	1.21e-08
1.21e-08	1.21e-08		1.21e-08			
0.20000000	0.198669	8.31e-04	3.31e-04	8.15e-05	1.76e-07	4.56e-08
4.56e-08	4.56e-08		4.56e-08			
0.30000000	0.295520	1.98e-03	4.88e-04	1.15e-04	4.54e-07	1.05e-07
1.05e-07	2.67e-07		1.05e-07			
0.40000000	0.389418	3.56e-03	6.29e-04	1.37e-04	8.27e-07	2.05e-07
3.52e-06	6.65e-07		2.11e-07			
0.50000000	0.479426	5.50e-03	7.46e-04	1.38e-04	1.13e-06	3.65e-07
6.49e-06	1.20e-06		5.30e-07			
0.60000000	0.564642	7.65e-03	8.31e-04	1.12e-04	1.06e-06	6.21e-07
9.06e-06	1.91e-06		8.36e-07			
0.70000000	0.644218	9.87e-03	8.79e-04	5.38e-05	1.66e-07	1.01e-06
1.10e-05	2.83e-06		1.11e-06			
0.80000000	0.717356	1.20e-02	8.88e-04	3.73e-05	2.07e-06	1.58e-06
1.22e-05	3.96e-06		1.34e-06			
0.90000000	0.783327	1.38e-02	8.61e-04	1.58e-04	6.09e-06	2.36e-06
1.28e-05	5.26e-06		1.50e-06			
1.00000000	0.841471	1.52e-02	8.04e-04	3.03e-04	1.21e-05	3.36e-06
1.27e-05	6.68e-06		1.58e-06			
1.10000000	0.891207	1.62e-02	7.27e-04	4.61e-04	2.02e-05	4.56e-06
1.21e-05	8.14e-06		1.60e-06			
1.20000000	0.932039	1.68e-02	6.37e-04	6.23e-04	2.99e-05	5.90e-06
1.11e-05	9.57e-06		1.55e-06			
1.30000000	0.963558	1.71e-02	5.43e-04	7.81e-04	4.05e-05	7.26e-06
9.86e-06	1.09e-05		1.45e-06			
1.40000000	0.985450	1.71e-02	4.49e-04	9.25e-04	5.13e-05	8.51e-06
8.52e-06	1.21e-05		1.30e-06			
1.50000000	0.997495	1.70e-02	3.59e-04	1.05e-03	6.13e-05	9.50e-06
7.16e-06	1.30e-05		1.12e-06			
1.60000000	0.999574	1.68e-02	2.75e-04	1.16e-03	6.98e-05	1.01e-05
5.82e-06	1.37e-05		9.20e-07			
1.70000000	0.991665	1.67e-02	1.96e-04	1.24e-03	7.61e-05	1.03e-05
4.54e-06	1.42e-05		7.10e-07			
1.80000000	0.973848	1.66e-02	1.22e-04	1.30e-03	8.01e-05	1.00e-05
3.32e-06	1.45e-05		4.95e-07			
1.90000000	0.946300	1.66e-02	5.10e-05	1.34e-03	8.17e-05	9.42e-06
2.13e-06	1.45e-05		2.81e-07			
2.00000000	0.909297	1.68e-02	1.82e-05	1.36e-03	8.10e-05	8.52e-06
9.55e-07	1.44e-05		7.02e-08			

Error table | h = 0.05 (N = 40)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
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Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	0.000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00	0.00e+00				
0.05000000	0.049979	2.08e-05	2.08e-05	5.21e-06	8.68e-10	1.35e-10
1.35e-10	1.35e-10	1.35e-10				
0.10000000	0.099833	1.04e-04	4.16e-05	1.04e-05	4.18e-09	5.97e-10
5.97e-10	5.97e-10	5.97e-10				
0.15000000	0.149438	2.49e-04	6.22e-05	1.54e-05	1.07e-08	1.39e-09
1.39e-09	7.83e-09	1.39e-09				
0.20000000	0.198669	4.56e-04	8.25e-05	2.03e-05	2.12e-08	2.55e-09
1.10e-07	1.66e-08	7.02e-09				
0.25000000	0.247404	7.22e-04	1.02e-04	2.47e-05	3.57e-08	4.13e-09
2.16e-07	2.59e-08	1.54e-08				
0.30000000	0.295520	1.04e-03	1.21e-04	2.86e-05	5.40e-08	6.20e-09
3.20e-07	3.62e-08	2.38e-08				
0.35000000	0.342898	1.42e-03	1.39e-04	3.16e-05	7.52e-08	8.88e-09
4.20e-07	4.77e-08	3.20e-08				
0.40000000	0.389418	1.84e-03	1.56e-04	3.36e-05	9.71e-08	1.23e-08
5.14e-07	6.08e-08	4.01e-08				
0.45000000	0.434966	2.31e-03	1.71e-04	3.44e-05	1.17e-07	1.67e-08
6.02e-07	7.59e-08	4.78e-08				
0.50000000	0.479426	2.80e-03	1.84e-04	3.36e-05	1.30e-07	2.23e-08
6.83e-07	9.34e-08	5.50e-08				
0.55000000	0.522687	3.32e-03	1.96e-04	3.11e-05	1.31e-07	2.93e-08
7.54e-07	1.14e-07	6.18e-08				
0.60000000	0.564642	3.86e-03	2.05e-04	2.68e-05	1.13e-07	3.80e-08
8.14e-07	1.37e-07	6.78e-08				
0.65000000	0.605186	4.39e-03	2.12e-04	2.05e-05	6.97e-08	4.88e-08
8.64e-07	1.62e-07	7.32e-08				
0.70000000	0.644218	4.92e-03	2.16e-04	1.22e-05	7.00e-09	6.18e-08
9.02e-07	1.92e-07	7.77e-08				
0.75000000	0.681639	5.44e-03	2.18e-04	1.90e-06	1.25e-07	7.75e-08
9.29e-07	2.24e-07	8.13e-08				
0.80000000	0.717356	5.92e-03	2.18e-04	1.03e-05	2.90e-07	9.61e-08
9.43e-07	2.58e-07	8.40e-08				
0.85000000	0.751280	6.37e-03	2.15e-04	2.43e-05	5.09e-07	1.18e-07
9.47e-07	2.95e-07	8.58e-08				
0.90000000	0.783327	6.78e-03	2.10e-04	3.98e-05	7.86e-07	1.43e-07
9.39e-07	3.35e-07	8.66e-08				
0.95000000	0.813416	7.14e-03	2.04e-04	5.66e-05	1.12e-06	1.71e-07
9.22e-07	3.75e-07	8.65e-08				
1.00000000	0.841471	7.46e-03	1.96e-04	7.44e-05	1.52e-06	2.02e-07
8.97e-07	4.17e-07	8.56e-08				
1.05000000	0.867423	7.72e-03	1.86e-04	9.30e-05	1.97e-06	2.36e-07
8.64e-07	4.59e-07	8.39e-08				
1.10000000	0.891207	7.94e-03	1.76e-04	1.12e-04	2.48e-06	2.72e-07
8.25e-07	5.01e-07	8.14e-08				
1.15000000	0.912764	8.10e-03	1.65e-04	1.31e-04	3.03e-06	3.11e-07
7.81e-07	5.43e-07	7.84e-08				
1.20000000	0.932039	8.23e-03	1.53e-04	1.50e-04	3.62e-06	3.50e-07

7.34e-07	5.83e-07	7.48e-08				
1.25000000	0.948985	8.32e-03	1.42e-04	1.68e-04	4.23e-06	3.90e-07
6.85e-07	6.22e-07	7.08e-08				
1.30000000	0.963558	8.38e-03	1.30e-04	1.86e-04	4.86e-06	4.28e-07
6.34e-07	6.58e-07	6.64e-08				
1.35000000	0.975723	8.41e-03	1.18e-04	2.03e-04	5.48e-06	4.66e-07
5.83e-07	6.92e-07	6.18e-08				
1.40000000	0.985450	8.43e-03	1.07e-04	2.19e-04	6.10e-06	5.00e-07
5.32e-07	7.23e-07	5.70e-08				
1.45000000	0.992713	8.42e-03	9.53e-05	2.34e-04	6.69e-06	5.30e-07
4.82e-07	7.51e-07	5.20e-08				
1.50000000	0.997495	8.41e-03	8.44e-05	2.47e-04	7.24e-06	5.56e-07
4.33e-07	7.76e-07	4.69e-08				
1.55000000	0.999784	8.39e-03	7.39e-05	2.59e-04	7.75e-06	5.77e-07
3.85e-07	7.98e-07	4.18e-08				
1.60000000	0.999574	8.37e-03	6.38e-05	2.70e-04	8.20e-06	5.92e-07
3.39e-07	8.17e-07	3.67e-08				
1.65000000	0.996865	8.36e-03	5.41e-05	2.80e-04	8.59e-06	6.00e-07
2.94e-07	8.32e-07	3.16e-08				
1.70000000	0.991665	8.34e-03	4.48e-05	2.88e-04	8.92e-06	6.02e-07
2.51e-07	8.44e-07	2.66e-08				
1.75000000	0.983986	8.34e-03	3.58e-05	2.95e-04	9.18e-06	5.98e-07
2.09e-07	8.53e-07	2.16e-08				
1.80000000	0.973848	8.34e-03	2.70e-05	3.01e-04	9.36e-06	5.88e-07
1.68e-07	8.59e-07	1.66e-08				
1.85000000	0.961275	8.36e-03	1.85e-05	3.06e-04	9.48e-06	5.73e-07
1.28e-07	8.62e-07	1.17e-08				
1.90000000	0.946300	8.39e-03	1.02e-05	3.09e-04	9.53e-06	5.53e-07
8.87e-08	8.62e-07	6.89e-09				
1.95000000	0.928960	8.43e-03	1.94e-06	3.12e-04	9.52e-06	5.28e-07
4.98e-08	8.60e-07	2.10e-09				
2.00000000	0.909297	8.48e-03	6.24e-06	3.14e-04	9.45e-06	5.01e-07
1.10e-08	8.56e-07	2.66e-09				